NEUROVA: 360° Multimodal Cognitive Learning & Diagnostic Engine

CODE KERNEL '26 · DECODE ROUND FINAL SPECIFICATION

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1. THE PROBLEM STATEMENT — WHY ONE-PACE INSTRUCTION FAILS INDIAN CLASSROOMS

In secondary STEM education across India (Classes 6 through 10 in CBSE, ICSE, and State curricula), the typical classroom contains **40 to 60 students managed by a single educator**. This environment creates a chronic structural failure known as **The Silent Indian Classroom Gap**:

1. The Single Fixed Speed Dilemma

Because the teacher must teach to the middle of the room, high-velocity learners wait in disengaged boredom while struggling students fall further behind, accumulating unaddressed gaps in foundational Class 6–7 mathematical concepts.

2. Compounding Latency Crisis

Traditional paper homework takes 3 to 7 days for manual teacher correction. Misconceptions in algebraic signs or geometric theorems calcify into permanent habits before end-of-chapter summative exams finally detect them.

3. Sidelining the School Teacher

Commercial coaching centers and generic video apps bypass the school teacher completely, turning homework into unverified rote copying and destroying the educator's diagnostic visibility into student logic.

★ **CORE MISSION SOLVED:** NEUROVA detects daily conceptual breakdowns in real-time through multimodal optical & speech AI while restoring the Indian school teacher to the authoritative center of the learning loop.

2. SOLUTION STEPS 1–4: ZERO-FRICTION ONBOARDING & THE 6-DIGIT CLASS CODE ARCHITECTURE

`STEPS 01-04`

Steps 1 & 2: Cover Gateway & Dual Role Split

Step 1 Orbital Cover: An intuitive, zero-clutter portal. Tapping the NEUROVA halo logo reveals core system highlights, About Us institutional credentials, and support footer.

Step 2 Dual Portals: A high-contrast role selection splits students and teachers into isolated environments. Built specifically for 11-to-16-year-old learners with instant 1-tap visual navigation and zero extraneous cognitive friction.

Steps 3 & 4: 6-Digit Class Code & Personal 4-Digit Security PIN

Step 3 Class Code Parsing: The student inputs a structured 6-digit identifier, e.g., `081236`:

- Digits 1–2 (`08`): Class / Grade Level (Class 8).
- Digit 3 (`1`): Class Section (`1 = Section A` , `2 = B` , `3 = C`).
- Digits 4–6 (`236`): Assigned Institutional Roll Number (#36).

Step 4 PIN Authentication & Profile Check: On first sign-in, the student sets a personal 4-digit PIN. The system validates against the pre-loaded school roster, displaying the student's exact name, section, and enrolled subjects.

★ **BHARAT-FIRST ADVANTAGE (STEPS 1–4):** In over 65% of Tier-2/Tier-3 Indian households, multiple siblings share a single family smartphone. The combination of the 6-digit Class Code and personal 4-digit PIN enables multiple siblings to safely log in on the same device without session cross-contamination or grade tampering.

3. SOLUTION STEPS 5–7: LONGITUDINAL ANALYSIS TELEMETRY & PERCENTILE CURVES

`STEPS 05-07`

Upon entry, NEUROVA presents the student with real-time longitudinal performance curves rather than static letter grades:

Step 5: Standing View

Immediately aggregates and displays where confirmed Student `081236` stands across all academic topics (Quadratic Equations, Optics, Cell Biology, Chemical Equations) relative to syllabus benchmarks.

Step 6: Real-Data Line

Plots empirical performance data derived from daily written homework accuracy, oral voice explanations, and follow-up re-tests over four progressive phases: *Wk 1 (Basics) → Wk 2 (Homework) → Wk 3 (Voice) → Wk 4 (Current)*.

Step 7: Class Cohort Overlay

Superimposes the student's personal trajectory line directly against the Class 8-A Cohort Average line (e.g. Student rising from 74% to 89% vs. Class average steady at 70%), providing healthy, non-punitive benchmark motivation.

4. SOLUTION STEPS 8 & 9: HYPER-GRANULAR SOLUTION AUDITING & STEP-LEVEL OCR

`STEPS 08-09`

Daily Task Assignment & Sub-Second Blur Gate

Sir's 5–10 Daily Tasks: Directly renders the 5 to 10 daily homework questions published by the school teacher, prompting the student to write out full calculations in their physical notebooks.

OpenCV.js Laplacian Blur Gate: To eliminate garbage inputs, the client-side camera module applies an on-device Laplacian variance kernel. If lighting is inadequate, ink is smudged, or camera shake causes blur, the upload is rejected in <500ms before cloud network transmission.

Deep Step-Level Multimodal AI OCR

Unlike commercial apps that only check the final number, NEUROVA's vision model audits **every intermediate line of mathematical derivation**:

- Extracts handwritten equations into structured LaTeX matrices.
- Distinguishes between algebraic slips, arithmetic miscalculations, and conceptual formula errors (e.g. missing denominator $2a$ vs. sign reversal).
- Pinpoints the precise step number where divergence occurred.

★ **KEY ADVANTAGE (STEPS 8–9):** Discovers the exact step where a student's cognitive model broke down, awarding step marks and preventing students from getting completely discouraged by an isolated calculation slip.

🧠 Cognitive Architecture: Cross-Modal Synthesis & Automated Remediation

SOLUTION STEPS 10 THROUGH 14 DEEP DIVE

5. SOLUTION STEP 10: NEURAL ACOUSTIC VOICE AI ENGINE & LIVE EDITABLE TRANSCRIPT

STEP 10

Written homework has a notorious vulnerability: students can copy steps from peers or online guides without understanding. NEUROVA introduces the **Oral Feynman Verification Protocol**:

Bilingual Acoustic AI Model

After uploading written notes, the student taps the microphone to explain their solution orally. The acoustic AI evaluates:

- **Semantic Conceptual Depth:** Verifies that core scientific terms (e.g., *"refraction index"*, *"discriminant"*, *"semipermeable membrane"*) are explained in context.
- **Vocal Confidence & Fluency:** Tracks hesitation pauses and tonal inflection to detect genuine understanding vs. memorized recitation.
- **Vernacular-to-English Bridging:** Recognizes bilingual scientific phrasing, supporting Indian English accents seamlessly.

Live Editable Real-Time Transcript

As the student speaks, an automated speech recognition pipeline streams an editable live transcript below the audio visualizer:

- Enables students to visually inspect what the speech engine transcribed.
- Allows manual editing to fix mispronounced or dialect-specific technical terms before final algorithmic evaluation.
- **Rote Defeat:** A student can copy written symbols, but cannot fake explaining the underlying conceptual logic aloud in their own voice.

★ **ORAL EXPLANATION ADVANTAGE:** Based on the Feynman Technique, verbally teaching a concept back to the system boosts long-term memory consolidation by 300% and instills English academic communication confidence.

6. SOLUTION STEP 11: CROSS-MODAL COGNITIVE FUSION & DIAGNOSTIC MATRIX

STEP 11

Dual-Engine Synthesis & Illusion Detection

NEUROVA executes cross-modal correlation by fusing **Step 9 Handwritten Derivations** with **Step 10 Semantic Voice Embeddings**:

- *Detects Hidden "Illusions of Competence":* Catches students who write flawless mathematical steps by mechanical memory but cannot articulate why the formula works when speaking.
- *Catches Expressive Thinkers:* Identifies students who make careless handwriting slips on paper but articulate perfect verbal mastery, ensuring fair diagnostic assessment.

Tri-Tier Visual Diagnostic Triage

Classifies every curricular sub-topic into three actionable diagnostic tiers with empirical score bars:

- **Strong Tiers (Score $ge 80%$):** e.g., *Quadratic Equations (94.0%)*, *Polynomials (86.0%)*. Marked as mastered; eligible for peer assistance.
- **Average Tiers ($50% ext{--}79%$):** e.g., *Triangles & Similarity (67.0%)*. Slated for targeted formula review.
- **Weak Tiers ($< 50%$):** e.g., *Trigonometric Ratios (36.0%)*. Automatically queues full multimodal remediation.

★ **FUSION MOAT (STEP 11):** Completely eliminates false positives in student assessment by requiring both procedural handwriting accuracy and conceptual verbal articulation simultaneously.

7. SOLUTION STEP 12: PERSONALIZED MULTIMODAL REMEDIATION & 10-FLASHCARD GENERATOR

STEP 12

Once conceptual weaknesses are isolated, the system launches an automated, three-part personalized learning intervention:

1. AI Voice Tutor (Audio)

Walks the student through missed steps using natural, paced speech. Features an interactive 3-speed selector (**1.0x Normal**, **1.25x Revision**, **1.5x Rapid**) and real-time karaoke text highlighting that centers the active spoken sentence.

2. Text Deep-Dive Notes

Delivers structured textbook-aligned revision summaries. Automatically highlights the exact theorems, sign conventions, and scientific principles where the student stumbled during the homework phase.

3. 10 Revision Flashcards

Synthesizes 10 interactive recall cards with 3D flip animations. Condenses core formulas, definition pairs, and diagram callouts into high-yield active retrieval cards for rapid self-testing before class examinations.

8. SOLUTION STEP 13: 3D VISUAL WORD SCENARIOS & REAL-WORLD APPLICATION LOOPS

STEP 13

Contextual Word Problem Generation

Traditional multiple-choice quizzes encourage blind guessing. Step 13 generates real-world word problems using the newly remediated concepts (e.g., calculating shadow angles for trigonometry or circuit resistance in household wiring).

Mental 3D Visualization & Repeat Voice Loop

Requires mental visualization of spatial physical scenarios instead of rote formula plugging. The student must once again explain the solution logic back to the AI by voice, proving closed-loop conceptual mastery.

9. SOLUTION STEP 14: CLOSED-LOOP VERIFICATION & TEACHER ESCALATION TRIGGER

STEP 14

Scenario A: If Solved

Mastery Unlocked: If the student solves the application scenario and passes the speech check, the topic is marked as mastered and their longitudinal curve updates dynamically.

Scenario B: If Still Weak

Gap Persistence Detected: If the student continues to struggle after automated AI reteaching, the engine identifies that human pedagogical intervention is required.

Scenario C: Flagged to Sir

Desk-Side Intervention Alert: Instantly pushes the student's profile and specific roadblock onto the school teacher's dashboard, prompting Sir to provide 1-on-1 guidance next morning.

★ HUMAN-IN-THE-LOOP SAFETY (STEP 14): Guarantees that zero students slip through the cracks by marrying algorithmic self-remediation with

10. TEACHER CONSOLE: COHORT TELEMETRY, WEAK-STUDENT GRAPHS & RETEACH TRIGGERS

NEUROVA is designed with a fundamental philosophy: **never replace or sideline the school teacher**. The Educator Suite provides teachers with actionable intelligence without adding administrative burden:

Evening Cohort Digest & X-Ray Vision

As students complete homework and voice tasks at home, NEUROVA compiles an **Evening Cohort Digest**. When Sir opens the portal:

- **Weak-Student Bar Graph:** Instantly sorts and displays struggling learners needing help (e.g. *Roll #12: 34%, Roll #29: 39%, Roll #41: 48%* against the *Class 8-A Average of 70%*).
- **1-on-1 Diagnostic Cards:** Sir can tap any student card to inspect the exact handwritten derivation errors and listen to their audio recordings.

The "Majority Reteach Rule" & Classroom Alerts

Teachers cannot manually grade 50 notebooks every evening. NEUROVA automates cohort diagnosis and triggers actionable alerts:

- **Majority Reteach Alert:** If $>50\%$ of the section fails a specific topic (e.g. *Sign convention in Mirror Formula*), Sir receives an alert: *"28/40 students failed Topic 4. Recommended: 10-minute classroom re-explanation."*
- **Desk-Side Focus:** Ranks students weak across multiple topics for personalized 2-minute morning check-ins.

★ **TEACHER ADVANTAGE:** Eliminates hours of manual red-pen notebook checking while giving Sir instant, high-resolution diagnostic visibility into an entire 50-student classroom.

11. OFFLINE CLASSROOM EXAM INTEGRATION: PAPER SLIP-TEST OCR & 100%/80% MASTERY BARS

Bridging Pen-and-Paper Exams with AI Analytics

Indian schools will continue to conduct pen-and-paper slip-tests, unit tests, and terminal exams for decades. NEUROVA requires zero change to this traditional workflow:

- Students write unit tests on physical paper question sheets.
- The teacher grades sheets using their traditional red pen.
- **Dual Scan Analysis:** The student or teacher snaps a photo of the question sheet and the teacher-marked answer paper.

Automated Sub-Topic Mapping & Mastery Bars

The vision OCR parses teacher checkmarks, deducted marks, and question headers, automatically converting physical marks into granular topic bars:

- *Quadratic Equations:* `100.0% Mastery [Strong]`
- *Polynomials:* `90.0% Mastery [Strong]`
- *Triangles:* `80.0% Mastery [Average]`
- *Trigonometry:* `40.0% Mastery [Weak - Intervention Triggered]`

★ **ZERO DISRUPTION ADVANTAGE:** Requires zero expensive smartboards or school tablets—bridges physical paper slip-tests into granular longitudinal analytics in under 10 seconds.

12. FULL-STACK ARCHITECTURAL TOPOLOGY & IMPLEMENTATION MATRIX

NEUROVA FEATURE	TECHNOLOGY UTILIZED	IMPLEMENTATION & OPERATIONAL MECHANISM
Cover & Dual Role Split	HTML5 + CSS3 + React 18	Renders cybernetic halo emblem, responsive portal selector, and zero-clutter navigation.
Class Code (081236) & PIN	JavaScript + Roster Store	Decodes Grade/Section/Roll #, enforces 4-digit PIN for multi-sibling phone sharing.
Blur & Ink Gate	OpenCV.js Laplacian Kernel	On-device sub-second blur detection, rejecting unusable camera captures prior to upload.
Handwritten Math OCR	Google Gemini Multimodal Vision	Parses multi-line derivations, separating sign slips from conceptual formula errors into LaTeX.
Speech & Live Transcript	Web Speech API + Whisper ASR	Evaluates oral semantic depth; streams real-time editable transcripts for pronunciation corrections.
Audio Teleprompter	Web Speech TTS (Chunk Engine)	Splits text into 180-char chunks to bypass Chromium GC bugs; synchronizes word-by-word karaoke ticker.
Visual Word Scenarios	Gemini Generative NLP Engine	Synthesizes contextual CBSE competency-based application word problems and 10 flashcards.
Telemetry Line/Bar Graphs	HTML5 Canvas API + SVG	Draws student vs. class cohort trajectory curves and 100%/80%/40% slip-test topic mastery bars.
Security Vault & Storage	Argon2id + localStorage Vault	Cryptographic password hashing, 5-attempt brute-force lockout, and client-side privacy isolation.

13. PRODUCTION DEPLOYMENT & INSTANT 1-CLICK EVALUATION CREDENTIALS

Student Portal (Class 10-D)

Code: `Class 10-D, Roll #21`
PIN: `pass`
Test: Solve daily tasks, submit handwritten notes, listen to 1.25x voice teleprompter, flip 10 flashcards.

Educator Suite (Dr. Evelyn Reed)

Email: `teacher@neurova.edu`
Password: `temp123`
Test: View weak student graphs, audit student derivations in zoom lightbox, review majority reteach alerts.

Institutional Console (Admin)

Email: `admin@neurova.edu`
Password: `MasterAdmin#2026`
2-Step OTP: `482910`
Test: Institutional rosters, audit security logs, review lockout triggers.

Live Production URL: https://neurova-seven-

GitHub: https://github.com/Anurag-

Team: Anurag, Siddarth, Solmon, Jithin